

Real Time Data Acquisition and analysis of Engine performance and vehicle Load using CAN protocol



PG-Diploma in Embedded Systems and Design (PG-DESD)

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Introduction

Modern vehicles demand real-time monitoring of performance data to ensure efficiency, safety, and predictive maintenance. CAN protocol is widely used in automotive systems due to its robustness and reliability in distributed sensor networks.

This project integrates multiple vehicle sensors using ESP32 microcontrollers communicating over the CAN bus. It allows vehicle parameters to be monitored locally on a TFT display and remotely via cloud through AWS using MQTT protocol. The system is modular and scalable, suitable for various transportation and fleet management applications.

Problem Statement

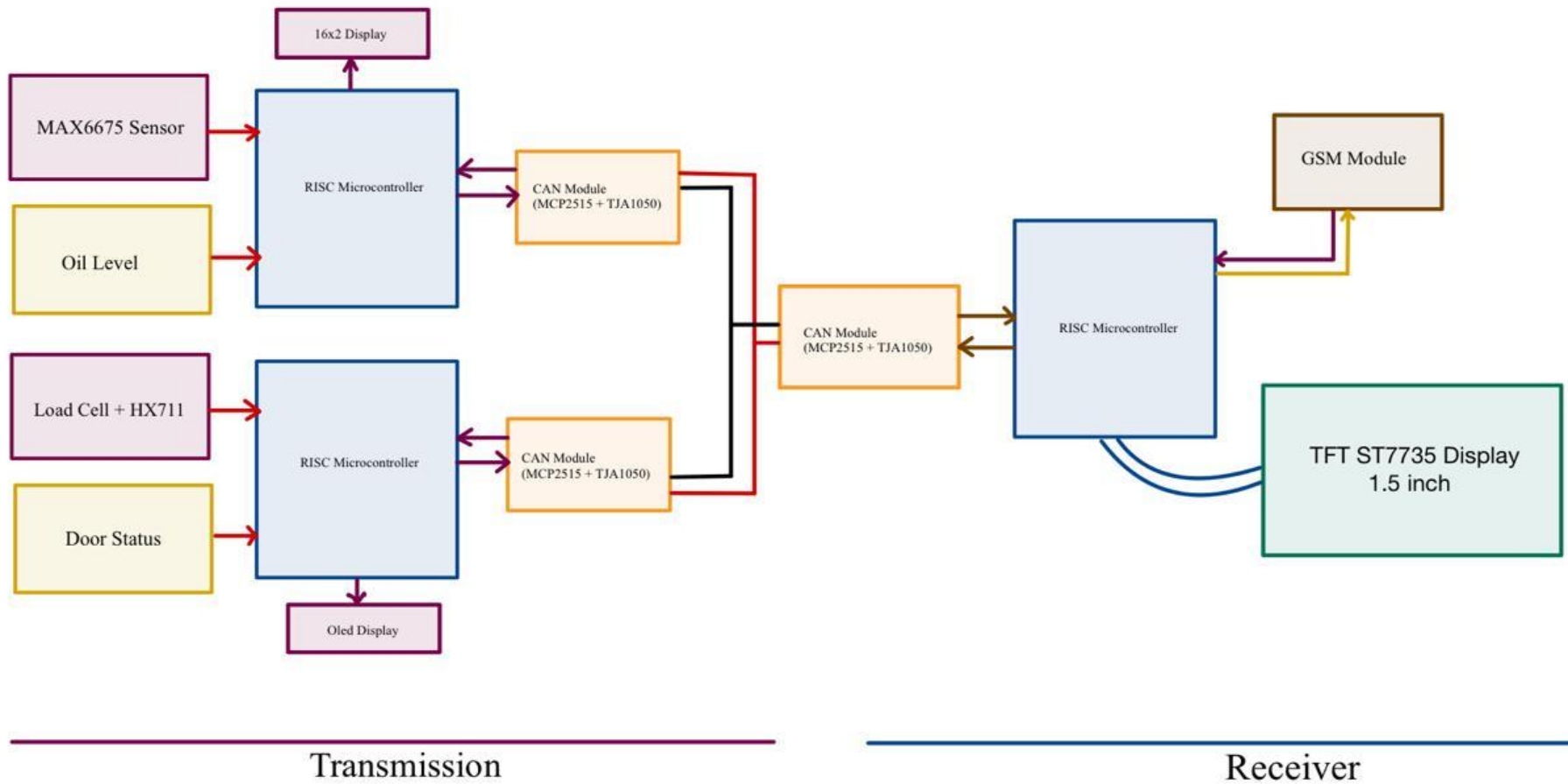
- Constant accident of heavy load vehicle.
- High cost fine and time to check load in vehicle.
- Lack of low-cost, scalable systems to collect multiple vehicle parameters in real-time.
- Existing systems often require wired logging without cloud access.
- Need for continuous monitoring even in remote areas.

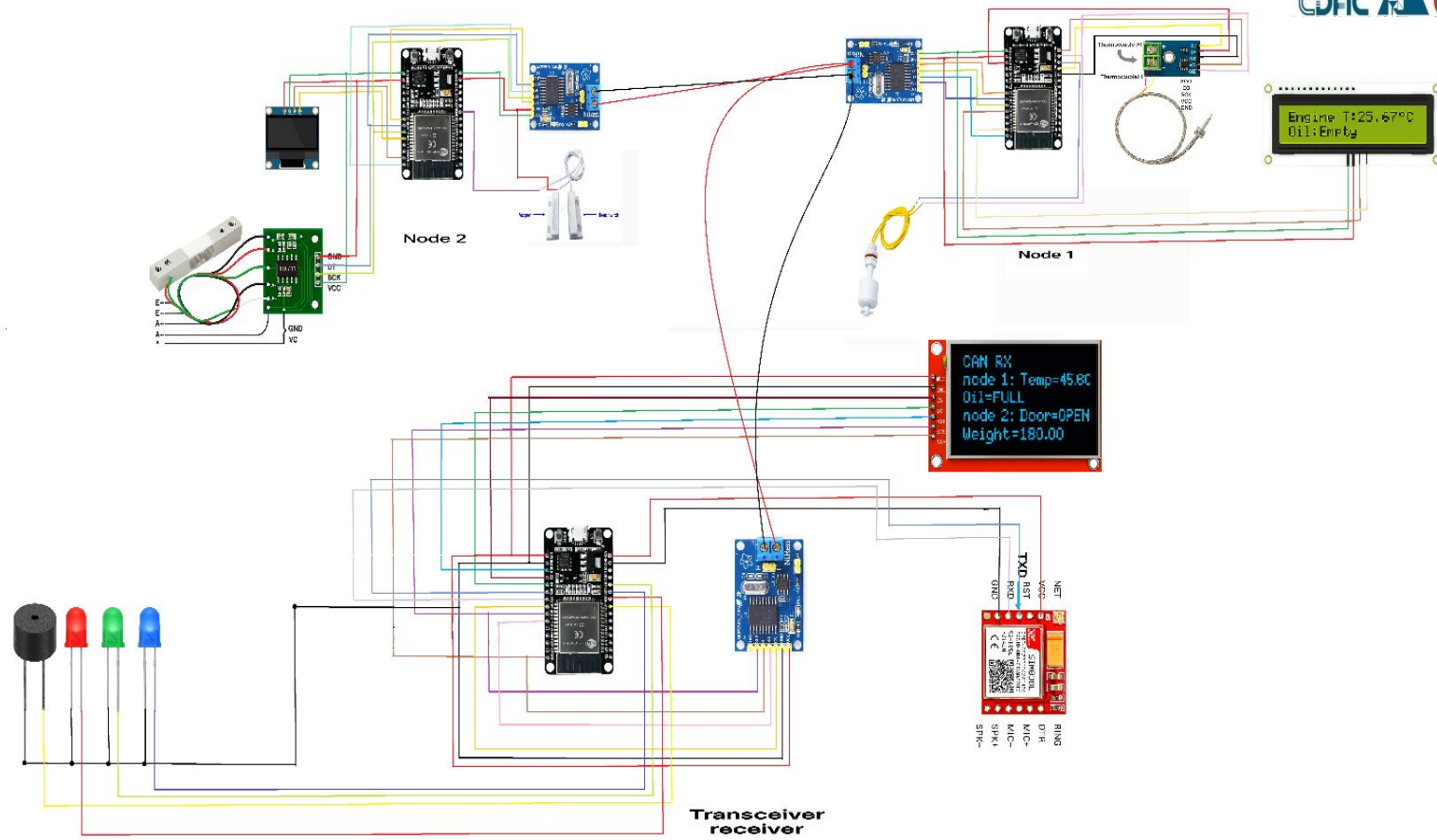
Objectives

- To design a multi-node CAN-based vehicle monitoring system
- To acquire temperature, fuel level, load and door status
- To visualize data locally on TFT
- To upload data to cloud (ThingSpeak)

Components Used (Hardware)

- ESP32-WROOM
- MCP2515 CAN Module
- MAX6575 Temperature Sensor
- Oil Level Sensor
- OLED Display
- HX711 + Load Cell
- Magnetic Reed Switch
- TFT Display
- SIM800L GSM Module





Transceiver receiver

Dashboard

Node 1: Temperature & Oil Level



0.0 °C

Oil Level:

Full



Connection Status

Node 2: Door Status, Load & GPS

Closed



0.00

Latitude: 12.971598

Longitude: 77.594566



Connection Status

Data Acquisition System

Channel ID: 3026763

Author: mwa0000017319566

Access: Private

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

+ Add Visualizations

+ Add Widgets

Export recent data

MATLAB Analysis

MATLAB Visualization

Channel 2 of 2 < >

Channel Stats

Created: a day ago

Last entry: about 3 hours ago

Entries: 335

Field 1 Chart

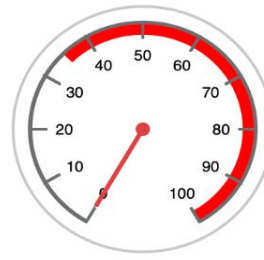


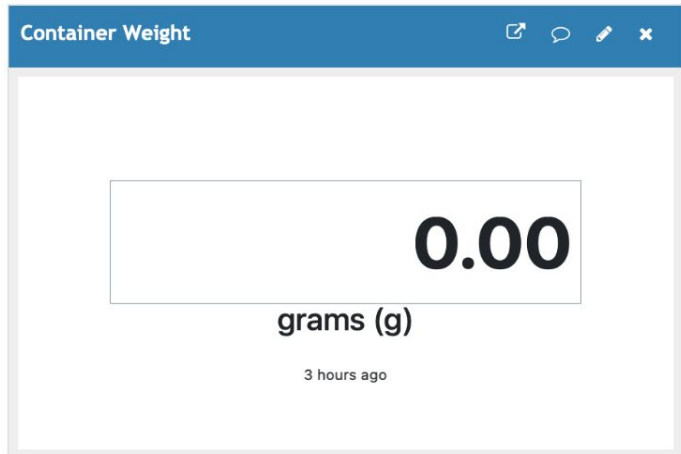
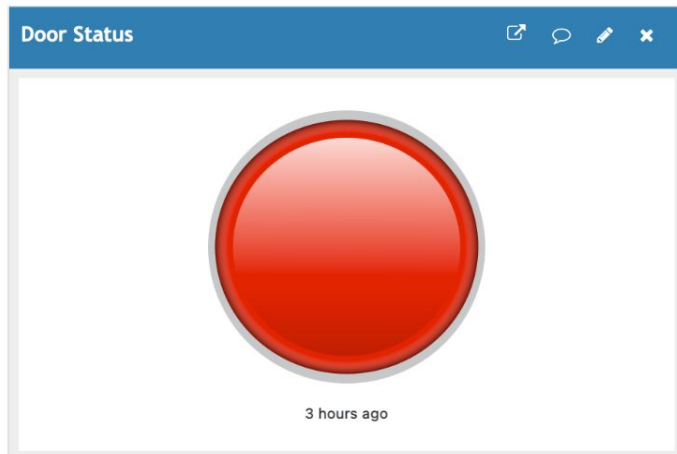
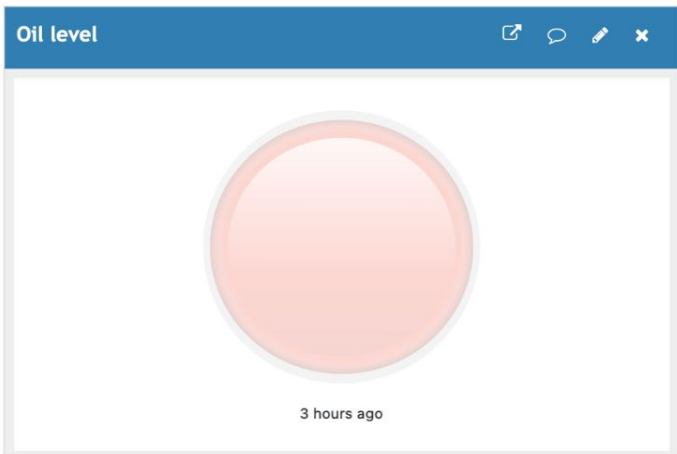
Data Acquisition System



ThingSpeak.com

Engine Temperature





Conclusion

- System provides reliable in-vehicle data acquisition
- Supports local and cloud monitoring
- Cost-effective and scalable design for smart vehicle applications

Future Scope

- Add more sensors (RPM, vibration, braking)
- Use AI for predictive maintenance
- Integration with vehicle ECU and mobile app
- Deploy in real vehicle fleets for field use

THANK YOU